

Zhicheng (Stark) Guo

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🔍 [Zhicheng Guo](#) (Google Scholar)

I develop **interpretable deep learning models** that empower clinicians to understand and trust AI-driven medical diagnostics. My work bridges the gap between **high-performance AI and clinical adoption** by keeping medical experts in the decision-making loop. My research focuses on **high-stakes** clinical applications including cardiac monitoring, EEG diagnostics, and breast cancer risk prediction, with work published in NEJM AI, Nature Machine Intelligence, CVPR, and JMLR.

EDUCATION Duke University

Durham, NC

Ph.D. in Electrical and Computer Engineering

Jan 2022 – May 2026

(Advisor: Cynthia Rudin)

M.S. in Computer Science (Transferred to Ph.D. Program)

Aug 2020 – Dec 2021

Rensselaer Polytechnic Institute

Troy, NY

B.Sc. in Computer Science

July 2016 – May 2020

PUBLICATIONS (* indicates co-first / senior authors, equal contribution)

Journal Publications

- [1] Varun Babbar*, **Zhicheng Guo***, Cynthia Rudin. "What is Different Between These Datasets? A Framework for Explaining Data Distribution Shifts"
Journal of Machine Learning Research (JMLR), 2025. 
- [2] Alina Jade Barnett*, **Zhicheng Guo***, Jin Jing*, Wendong Ge, Brandon Westover, Cynthia Rudin. "Improving Clinician Performance in Classification of EEG Patterns on the Ictal-Interictal-Injury Continuum using Interpretable Machine Learning."
New England Journal of Medicine AI (NEJM AI), 2024. 
- [3] **Zhicheng Guo**, Cheng Ding, Duc H. Do, Amit Shah, Randall J Lee, Xiao Hu, Cynthia Rudin. "SiamAF: Improving atrial fibrillation detection using a shared latent space for ECG and PPG signals."
Harvard Data Science Review (HDSR), 2024. 
- [4] Cheng Ding, **Zhicheng Guo**, Cynthia Rudin, Ran Xiao, Amit Shah, Duc H Do, Randall J Lee, Gari Clifford, Fadi B Nahab, Xiao Hu. "Learning From Alarms: A Robust Learning Approach for Accurate Photoplethysmography-Based Atrial Fibrillation Detection Using Eight Million Samples Labeled with Imprecise Arrhythmia Alarms."
IEEE Journal of Biomedical and Health Informatics (JBHI), 2024. 
- [5] Sully F Chen, **Zhicheng Guo**, Cheng Ding, Xiao Hu, Cynthia Rudin. "Learned Kernels for Sparse, Interpretable, and Efficient Medical Time Series Processing."
Nature Machine Intelligence, 2024. 
- [6] Cheng Ding, **Zhicheng Guo**, Zhaoliang Chen, Randall J Lee, Cynthia Rudin, Xiao Hu. "SiamQuality: a ConvNet-based Foundation Model for Photoplethysmography Signals."

Physiological Measurement, 2024. 

- [7] Manickam Ashokkumar, Wenwen Mei, Jackson J Peterson, Yuriko Harigaya, David M Murdoch, David M Margolis, Caleb Kornfein, Alex Oesterling, **Zhicheng Guo**, Cynthia D Rudin, Yuchao Jiang, Edward P Browne. "Integrated Single-cell Multiomic Analysis of HIV Latency Reversal Reveals Novel Regulators of Viral Reactivation."

Genomics, Proteomics & Bioinformatics, 2024. 

- [8] **Zhicheng Guo**, Cheng Ding, Xiao Hu, Cynthia Rudin. "A Supervised Machine Learning Semantic Segmentation Approach for Detecting Artifacts in Plethysmography Signals from wearables."

Physiological Measurement, 2021. 

Conference Publications

- [9] Jon Donnelly, **Zhicheng Guo**, Alina Jade Barnett, Hayden McTavish, Chaofan Chen, Cynthia Rudin. "Rashomon Sets for Prototypical-Part Networks: Editing Interpretable Models in Real-Time."

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. 

- [10] Ruixin Zhang, Jon Donnelly, **Zhicheng Guo**, Ghazal Khalighinejad, Haiyang Huang, Alina Jade Barnett, Cynthia Rudin. "It's LIT! LLMs with Interpretable Tools."

Multi-Turn Interactions in Large Language Models Workshop (MTI-LLM, NeurIPS Workshop), 2025. [Accepted]

Submitted

- [11] Anish Karpurapu, **Zhicheng Guo**, Xiao Hu, Cynthia Rudin "The Unverifiable Promises of Proprietary Health AI Models." *NPJ AI*, 2025.

PROFESSIONAL EXPERIENCE

Texas Instruments – Kilby Labs

May 2025 – Aug 2025

Machine Learning Researcher Intern

- Developed an **LLM agentic framework** for hardware-constrained neural architecture and **mixed-precision quantization** co-optimization.
- Implemented multi-objective reward functions and on-the-fly hardware profiling to guide optimization decisions and improve real-time performance.
- Developed specialized small reasoning language model groups (SLMs) as replacements for generic **LLM agents** with **GRPO and finetuning**, enabling on-site and data-secure inference at a fraction of compute cost.

Duke University Health System

May 2021 – Aug 2021

Quantitative Research Intern

- Investigated correlations between cellular populations/markers in CSF and PBMC with cognitive scores in HIV-associated neurocognitive disorder patients under different drug use conditions.
- Developed machine learning models with flow cytometry data to predict drug use and impairment, enabling clinical researchers to assess drug effects on patients.

PATENTS

Texas Instruments – Kilby Labs

- [1] LLM-guided neural network architecture and quantization search for TINIE (Texas Instruments Neural Inference Engine) on microcontrollers. (pending)
- [2] Small language models for TINIE-optimized edge AI model development. (pending)

MEDIA OP-EDS

- [1] Cynthia Rudin, **Zhicheng Guo**, Cheng Ding, and Xiao Hu. "How good are AI health technologies? We have no idea." *STAT News*, Oct. 11, 2023. 

News Articles

- [2] "AI helps medical professionals read confusing brain EEGs" *Mindplex*, May 29, 2024. 
- [3] "New AI tool from Duke researchers reads EEGs, could improve treatment of seizures" *Yahoo News (The News&Observer)*, Feb. 20, 2025. 

ACADEMIC SERVICES

Reviewer

IEEE Journal of Biomedical and Health Informatics (JBHI)	2023, 2024, 2025
Physiological Measurement	2025
Epilepsia	2024
NeurIPS	2024, 2025

TEACHING Duke University

Graduate Theory and Algorithms for Machine Learning 671D	Fall 2025
- Lecture on neural networks and Interpretable Neural Networks Design	
TA, Graduate Theory and Algorithms for Machine Learning 671D	Fall 2023
TA, Graduate Theory and Algorithms for Machine Learning 671D	Fall 2022
TA, Graduate Theory and Algorithms for Machine Learning 671D	Fall 2021

Rensselaer Polytechnic Institute

SAPRIS Program Academic Mentor	Summer 2017
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INVITED TALKS INFORMS

Interpretable Deep Learning Models	Atlanta, GA October 2025
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ADVISED STUDENTS

Will Li
 - Master's student, at Duke University
 Anish Karpurapu
 - MD, PhD student at Duke University
 Maximillian Machado
 - Master student, incoming PhD student at Duke University
 Liam Connor
 - Master student at Duke University
 Pranay Jain
 - Former master's student, now PhD student at University of Washington
 Haoyu (Cindy) Wu
 - Former master's student, now in industry
 Dennis Tang

- Undergraduate student, incoming PhD student at Duke University

Alex Oesterling

- Former undergraduate student, now PhD student at Harvard University

Edna Zhang

- Former master's student, now in industry

Adam Costarino

- Former master's student, now in industry

Bhrij Patel

- Former undergraduate student, now PhD student at University Maryland